



Exercise

1. Encircle the correct answer:

- i) A digital signal has:
(a) Infinite values (b) Only two values
(c) Three values (d) Ten values
- ii) Which gate gives output 1 only when all inputs are 1?
(a) OR gate (b) AND gate
(c) NOT gate (d) NOR gate
- iii) XOR gate gives output 1 when inputs are:
(a) Same (b) Different
(c) Both 0 (d) Both 1
- iv) Which phase comes first in SDLC?
(a) Testing (b) Requirement Analysis
(c) Coding (d) Design
- v) In Agile, a short development cycle is called:
(a) Phase (b) Sprint
(c) Module (d) Release
- vi) Which testing type does NOT require knowledge of code?
(a) White box testing (b) Black box testing
(c) Unit testing (d) Code review
- vii) What device operates at the Network layer (Layer 3) of the OSI model?
(a) Hub (b) Switch
(c) Router (d) Repeater
- viii) How many layers does the TCP/IP model have?
(a) 3 (b) 4
(c) 5 (d) 7
- ix) UDP is preferred over TCP for video streaming because:
(a) UDP is more reliable (b) UDP is slower but more accurate
(c) UDP is faster and has lower delay (d) UDP guarantees delivery
- x) Which OSI layer is responsible for routing packets between networks?
(a) Physical (b) Data Link
(c) Network (d) Transport

2. Fill in the Blanks:

- i) An _____ gate gives output 1 only when both the inputs are same.
- ii) The _____ gate is also called an inverter.
- iii) In Agile, a short development cycle is called a _____.

- ◆ Searching is the process of locating a specific element within a collection of data.
- ◆ Common searching algorithms include:
 - ◆ Linear Search
 - ◆ Binary Search
- ◆ Correctness is the property of an algorithm that ensures it produces the expected output for every valid input.
- ◆ Efficiency of an algorithm is measured by the amount of time it takes to execute (time complexity) and the amount of memory it uses (space complexity).
- ◆ Clarity refers to how easy an algorithm is to read, understand, and modify by others.



1. Encircle the correct answer

- i) Which strategy involves ignoring the brand of a kettle when learning how to make tea?
 - (a) Decomposition
 - (b) Pattern Recognition
 - (c) Abstraction
 - (d) Sorting
- ii) Bubble Sort is also commonly known as:
 - (a) Linear Sort
 - (b) Ripple Sort
 - (c) Selection Sort
 - (d) Binary Sort
- iii) Which searching algorithm requires the data to be sorted before it can be used?
 - (a) Linear Search
 - (b) Sequential Search
 - (c) Binary Search
 - (d) Bubble Search
- iv) What is the main disadvantage of Selection Sort?
 - (a) It uses extra RAM
 - (b) It is difficult to code
 - (c) It must finish every loop even if the list is sorted
 - (d) It only works on alphabetical data
- v) An algorithm is considered "correct" if:
 - (a) It runs very fast
 - (b) Its output matches expected outcomes for given inputs
 - (c) It uses minimal memory
 - (d) It is written in a specific programming language
- vi) What does the abbreviation CT stand for?
 - (a) Computer Technology
 - (b) Computational Thinking
 - (c) Central Terminal
 - (d) Coding Technique
- vii) Which operation is represented by MOD?
 - (a) Multiplication
 - (b) Division
 - (c) Subtraction
 - (d) Finding the remainder after division

- viii) In algorithms, what is the purpose of a PTR?
- (a) To store text data (b) To perform calculations
(c) To track positions in an array (d) To display output
- ix) What does INT represent in an algorithm?
- (a) Internet (b) Integer
(c) Interval (d) Input
- x) In the Binary Search algorithm, the abbreviations BEG, END, and MID represent:
- (a) Input, Process, and Output (b) Beginning, End, and Middle positions
(c) First, Last, and Next values (d) Start, Stop, and Result

2. Fill in the Blanks

- i) _____ is a step-by-step logical procedure to solve a problem.
- ii) Sorting may be Ascending or _____.
- iii) Algorithm is _____ and language-independent.
- iv) _____ algorithm is designed to locate a specific element within list of data.
- v) We measure any algorithm efficiency through _____ and _____ as resources.

3. Provide short answers to the following questions.

- i) Define pattern recognition.
- ii) Explain the concept of decomposition.
- iii) What is meant by correctness in problem-solving?
- iv) Why is efficiency important in problem-solving?
- v) What do you understand by clarity in problem-solving?

4. Provide detailed answers to the following questions.

- i) What is the difference between an algorithm and pseudocode?
- ii) What is abstraction in an algorithm, and why is it used?
- iii) Write the steps of the Selection Sort algorithm.
- iv) Write the steps of the Binary Search algorithm.
- v) Which searching algorithm is best for large datasets? Explain your answer.



Activities

- i) Arrange the assembly queue according to students' heights.
- ii) Arrange the names in the attendance register in descending alphabetical order.
- iii) Find the name of your class monitor in the attendance register using the binary search technique.



Exercise

1. Encircle the correct answer:

- i) Which is the correct statement to insert Comments in Python code?
 - (a) // This is comment
 - (b) /* This is comment*/
 - (c) # This is comment
 - (d) - - This is comment
- ii) Which one is not a legal variable name?
 - (a) _myvar
 - (b) my_var
 - (c) Myvar
 - (d) my-var
- iii) Which one is the correct way to write if statement in Python?
 - (a) if (condition):
 - (b) if(condition);
 - (c) if (1):
 - (d) none of these
- iv) ____ is the operator can be used to compare two values?
 - (a) < >
 - (b) ==
 - (c) ><
 - (d) =
- v) ____ loop is used when the number of iterations is known in advance
 - (a) While
 - (b) Do while
 - (c) For
 - (d) For each
- vi) Which programming language is closest to binary (0s and 1s)?
 - (a) High Level language
 - (b) Mid-Level Language
 - (c) Low Level language
 - (d) Script Language
- vii) Standard indentation in Python?
 - (a) 2 spaces
 - (b) 4 spaces
 - (c) 6 spaces
 - (d) 8 spaces
- viii) ____ data type does the input() function always return.
 - (a) Int
 - (b) Float
 - (c) String
 - (d) Bool
- ix) ____ statement exits a loop immediately
 - (a) Pass
 - (b) Continue
 - (c) Break
 - (d) Exit
- x) ____ bitwise operator returns 1 only when BOTH bits are 1?
 - (a) AND
 - (b) OR
 - (c) NOT
 - (d) XOR

- ◆ A database is composed of tables, which store data in rows and columns.
- ◆ Each row is a record representing a single entity instance, and each column is a field representing an attribute.
- ◆ The primary key is used to uniquely identify each record, ensuring accuracy and preventing duplication.
- ◆ The Entity-Relationship (ER) Model provides a way to represent real-world systems using entities (objects), attributes (properties), and relationships (associations).
- ◆ ER diagrams visually illustrate how data elements are connected.
- ◆ Relationships can be:
 - One-to-One (1:1)
 - One-to-Many (1:M)
 - Many-to-Many (M:N) (resolved using a junction table)
- ◆ To maintain correctness across related tables, the concept of referential integrity is introduced.
- ◆ Referential integrity ensures that relationships remain valid by enforcing rules on foreign keys.
- ◆ Mechanisms such as cascade update and cascade delete automatically maintain consistency across tables.
- ◆ A relational schema acts as a blueprint of the database.
- ◆ Forms provide a user-friendly interface for entering, updating, and viewing data without directly interacting with tables.
- ◆ Queries are used to retrieve, filter, and analyze specific data from one or more tables based on defined criteria.
- ◆ Reports are used to present data in a formatted and organized manner, suitable for printing and sharing.



Exercise

- 1. Encircle the correct answer:**
- i) Which of the following is a DBMS?
 - (a) MS Word
 - (b) MS Excel
 - (c) MS Access
 - (d) MS Paint
 - ii) In an ER Model, a rectangle represents:
 - (a) Attribute
 - (b) Relationship
 - (c) Entity
 - (d) Key
 - iii) A one-to-many relationship means
 - (a) One record relates to many
 - (b) Many to one only
 - (c) One to One
 - (d) None

- iv) Which object is used to retrieve data?
 - (a) Table
 - (b) Form
 - (c) Query
 - (d) Report
- v) Which function calculates average?
 - (a) Sum()
 - (b) Count()
 - (c) Avg()
 - (d) Max()
- vi) Which key uniquely identifies each record in a table?
 - (a) Foreign Key
 - (b) Primary Key
 - (c) Alternate Key
 - (d) Candidate Key
- vii) Which relationship requires a junction table?
 - (a) One-to-One
 - (b) One-to-Many
 - (c) Many-to-Many
 - (d) None
- viii) Which database objects provides a user friendly interface for data entry?
 - (a) Query
 - (b) Table
 - (c) Report
 - (d) Form
- ix) Which chart is best used to show trends over time?
 - (a) Pie Chart
 - (b) Bar Chart
 - (c) Line Chart
 - (d) Column Chart
- x) Which query operation combines records into categories for summarization?
 - (a) Sorting
 - (b) Filtering
 - (c) Group by
 - (d) Validation

2. Fill in the Blanks:

- i) In a relational database, a _____ key is used to establish a link between two tables by referring to the primary key of another table.
- ii) If two entities are connected through an intermediate entity instead of directly, this usually indicates the resolution of a _____ relationship.
- iii) When a record in one table is deleted, and related records in another table must also be removed automatically, this is controlled through _____ rules.
- iv) A table created to resolve a many-to-many relationship is commonly referred to as a _____ table.
- v) A candidate key that is not chosen as the primary key is known as _____.

3. Provide short descriptive answers of the following questions.

- i) Differentiate between a field and a record.
- ii) What is referential integrity?
- iii) Explain the role of a Database Management System (DBMS).
- iv) Explain a one-to-many relationship with an example.
- v) Define data visualization and give one example.

- ◆ The main components of IoT are sensors, actuators, connectivity, data processing, cloud computing, and user interface.
- ◆ IoT is implemented in smart cities, industry, healthcare, education, smart homes, and agriculture.
- ◆ In Pakistan, IoT can enhance traffic management, healthcare services, classroom administration, industrial production, and water-efficient agriculture.
- ◆ Data Analytics involves collecting, storing, cleaning, analyzing, and presenting data to extract valuable insights.
- ◆ Data Analytics supports schools, banks, hospitals, businesses, and government offices in making informed decisions.
- ◆ Information sources are divided into primary, secondary, and tertiary sources.
- ◆ Students should check information carefully before using it because not all online information is reliable.
- ◆ Computing has positive impacts, such as faster communication, easier learning, better healthcare, online banking, and improved farming.
- ◆ Computing also has negative impacts, including privacy risks, excessive screen time, job displacement, and overreliance on technology.
- ◆ Responsible use of computing means using technology safely, ethically, and wisely.
- ◆ Assistive technologies support individuals with disabilities in mobility, learning, communication, hearing, vision, and daily activities.
- ◆ These technologies promote equal access to education, healthcare, employment, and social participation.
- ◆ This unit helps students understand the connection between computing, daily life, society, career opportunities, and Pakistan's future development.

Exercise

- 1. Encircle the correct answer**
- i) A smartwatch records heart rate and sends it to a mobile phone. Which technology is mainly used here?
 - (a) Word Processing
 - (b) Internet of Things
 - (c) Printing Technology
 - (d) Analog Technology
 - ii) In a smart irrigation system, which device checks whether the soil is dry or wet?
 - (a) Actuator
 - (b) Sensor
 - (c) Monitor
 - (d) Smart hearing aid

- iii) A school removes duplicate student records while preparing examination results. Which step of Data Analytics is this?
 - (a) Data collection
 - (b) Data visualization
 - (c) Data cleaning
 - (d) Data storage
- iv) Which of the following is a primary source of information?
 - (a) Textbook
 - (b) Interview
 - (c) Dictionary
 - (d) Summary website
- v) A student uses an unknown website without an author name or references. This source is likely to be:
 - (a) Reliable
 - (b) Primary
 - (c) Unreliable
 - (d) Official
- vi) Which technology supports people with hearing, vision, learning, or movement difficulties?
 - (a) Gaming technology
 - (b) Manufacturing technology
 - (c) Assistive technology
 - (d) Antivirus technology
- vii) Which of the following is an actuator in a smart water system?
 - (a) Soil moisture sensor
 - (b) Water valve
 - (c) Temperature sensor
 - (d) Camera sensor
- viii) Which type of source usually provides summarized information for quick reference?
 - (a) Primary source
 - (b) Secondary source
 - (c) Tertiary source
 - (d) Unverified source
- ix) Which method is used to present data through charts and graphs?
 - (a) Data collection
 - (b) Data storage
 - (c) Data cleaning
 - (d) Data visualization
- x) Which assistive technology converts on-screen text into speech?
 - (a) Screen reader
 - (b) Smart meter
 - (c) Motion sensor
 - (d) Water pump

2. Fill in the Blanks

- i) A _____ detects changes such as temperature, movement, light, or soil moisture.
- ii) An _____ performs a physical action, such as opening a valve or switching on a fan.
- iii) Correcting duplicate, missing, or incorrect records is called data _____.
- iv) An interview is an example of a _____ source of information.
- v) Screen readers and hearing aids are examples of _____ technologies.



Summary

- ◆ Digital literacy involves searching carefully, checking source reliability, collecting evidence, and communicating findings responsibly.
- ◆ A strong project begins with a focused research question and a suitable plan for using primary and/or secondary data.
- ◆ Interviews, surveys, prototypes, simulations, and observation each produce different kinds of evidence and should be chosen thoughtfully.
- ◆ Sampling, bias, privacy, consent, and citation all affect the quality and ethics of student research.
- ◆ Data should be cleaned and organized before analysis so that tables, charts, and conclusions are accurate.
- ◆ Different visuals serve different purposes: tables show exact values, bar charts compare categories, line graphs show trends, and infographics summarize key findings quickly.
- ◆ A strong digital artefact connects the research question, method, findings, conclusion, and references in a clear structure.
- ◆ Results should be communicated differently for different audiences while remaining accurate, readable, and evidence-based.



Exercise

1. Encircle the correct answer

- i) Which of the following best describes digital literacy?
 - (a) The ability to type quickly using a keyboard
 - (b) The ability to use digital tools to find, evaluate, create, and communicate information responsibly
 - (c) The ability to install software on a computer
 - (d) The ability to use social media for entertainment
- ii) A student asks classmates, “How does mobile phone use affect your concentration during study?” What type of data is mainly collected?
 - (a) Quantitative data because it can be counted easily
 - (b) Secondary data because classmates already know the answer
 - (c) Qualitative data because it collects opinions and experiences
 - (d) Numerical data because mobile phone use is related to time

- iii) Which search query is most suitable for finding a reliable PDF report about internet usage in Pakistan?
 - (a) internet usage Pakistan
 - (b) best internet use in Pakistan social media
 - (c) "internet usage in Pakistan" filetype:pdf
 - (d) Pakistan internet all websites latest
- iv) Which option is the best example of primary data?
 - (a) A government report downloaded from the internet
 - (b) A research article written by another author
 - (c) Survey responses collected directly from Class XI students
 - (d) Statistics copied from an online database
- v) Why are spreadsheets useful when presenting research data?
 - (a) They automatically make the research question correct
 - (b) They help organize data, perform calculations, and create charts
 - (c) They replace the need for collecting original data
 - (d) They ensure all online information is reliable
- vi) A digital artefact created after research should mainly:
 - (a) repeat long paragraphs from websites
 - (b) communicate results and conclusions clearly
 - (c) include decoration only
 - (d) avoid any data visuals
- vii) Data collected directly by the researcher is called:
 - (a) Secondary data
 - (b) Processed data
 - (c) Primary data
 - (d) Digital data
- viii) Which search operator helps search within a specific website or domain?
 - (a) OR
 - (b) AND
 - (c) NOT
 - (d) site:
- ix) Copying someone else's work without giving credit is called:
 - (a) Copyright
 - (b) Referencing
 - (c) Citation
 - (d) Plagiarism
- x) Asking permission before collecting responses is part of research:
 - (a) Analysis
 - (b) Sampling
 - (c) Presentation
 - (d) Ethics

2. Fill in the Blanks

- i) A well-defined _____ helps a student choose suitable data-collection methods and stay focused during research.
- ii) Data collected directly by the researcher through surveys, interviews or observation is called _____ data.